

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
NAME

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GROUP

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INDEX
NUMBER

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CHEMISTRY

9647/03

Paper 3 Free Response

22 September 2015

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Data Booklet
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.
A Data Booklet is provided.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

Answer any **four** questions

1 Chlorine undergoes the following equilibrium in aqueous solution.



- (a) (i) Write an expression for the equilibrium constant for this reaction, K_c , stating its units. [1]

$$K_c = \frac{[\text{Cl}^-][\text{HOCl}][\text{H}^+]}{[\text{Cl}_2]} \quad \text{mol}^2 \text{dm}^{-6}$$

- (ii) When 33 g of Cl_2 was bubbled into 100 cm^3 of water, and allowed to reach equilibrium, the resulting $[\text{Cl}_2(\text{aq})] = 4.57 \text{ mol dm}^{-3}$.

Calculate the equilibrium values of $[\text{Cl}^-(\text{aq})]$ and $[\text{HOCl}(\text{aq})]$ in this solution and hence calculate a value for K_c . [2]

Amount of $\text{Cl}_2 = 33 / 71 = 0.4648 \text{ mol}$

initial $[\text{Cl}_2] = 0.4648 / 0.1 = 4.648 \text{ mol dm}^{-3}$

	$\text{Cl}_2(\text{aq})$	$+$	$\text{H}_2\text{O}(\text{l})$	$\rightleftharpoons$	$\text{Cl}^-(\text{aq})$	$+$	$\text{HOCl}(\text{aq})$	$+$	$\text{H}^+(\text{aq})$
Initial [] / mol dm^{-3}	4.648				0		0		0
Δ in [] / mol dm^{-3}	-x				+x		+x		+x
Eqm [] / mol dm^{-3}	$4.648 - x$				x		x		x

$$4.648 - x = 4.57$$

$$x = 0.078 \text{ mol dm}^{-3} = 0.078 \text{ mol dm}^{-3}$$

$$[\text{Cl}^-] = [\text{HOCl}] = [\text{H}^+] = 0.078 \text{ mol dm}^{-3}$$

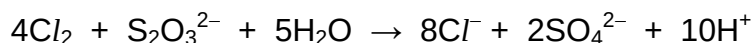
$$K_c = \frac{(0.078)^3}{(4.648 - 0.078)} = 1.04 \times 10^{-4} \text{ mol}^2 \text{dm}^{-6} \quad (3 \text{ s.f.})$$

- (iii) Describe and explain the effect of adding water to *equilibrium 1* above. [2]

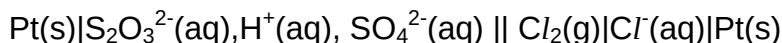
When water is added, $[\text{Cl}_2]$, $[\text{Cl}^-]$, $[\text{HOCl}]$ and $[\text{H}^+]$ decreases.

The side with more number of moles suffers a greater extent of dilution / numerator in K_c decreases more than denominator. Hence equilibrium will shift to the right to produce more Cl^- , HOCl and H^+ , in order to keep K_c constant.

- (b) Chlorine is a strong oxidising agent which can oxidise thiosulfate ion readily to sulfate in the following overall equation.



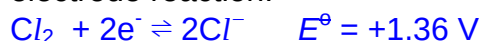
The cell consisting of a $\text{SO}_4^{2-}(\text{aq}), \text{H}^+(\text{aq}), \text{S}_2\text{O}_3^{2-}(\text{aq})|\text{Pt}(\text{s})$ half-cell and a $\text{Cl}_2(\text{g})|\text{Cl}^-(\text{aq})|\text{Pt}(\text{s})$ half-cell was set up, and shown using conventional cell notation below.



- (i) Construct an equation for the anode reaction. [1]



- (ii) The cell is capable of producing an e.m.f of +1.04V. By using suitable data from the *Data Booklet*, calculate a value for the $E^\ominus$ of the $\text{SO}_4^{2-}/\text{S}_2\text{O}_3^{2-}$ electrode reaction. [2]



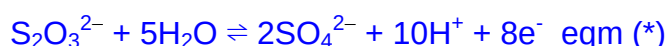
$$E^\ominus_{\text{cell}} = E^\ominus_{\text{red}} + E^\ominus_{\text{ox}}$$

$$1.04 = 1.36 + E^\ominus_{\text{ox}}$$

$$E^\ominus_{\text{ox}} = -0.32 \text{ V}$$

$$E^\ominus = +0.32 \text{ V}$$

- (iii) Explain how addition of $\text{Ba}(\text{NO}_3)_2(\text{aq})$ to the $\text{SO}_4^{2-}(\text{aq}), \text{H}^+(\text{aq}), \text{S}_2\text{O}_3^{2-}(\text{aq})|\text{Pt}(\text{s})$ half-cell will affect the e.m.f. of the cell. [2]



Addition of $\text{Ba}(\text{NO}_3)_2$ causes SO_4^{2-} ions to precipitate as $\text{BaSO}_4(\text{s})$. This decreases the $[\text{SO}_4^{2-}]$, resulting in the equilibrium (*) shifting to the right, / favouring the oxidation of $\text{S}_2\text{O}_3^{2-}/E_{\text{ox}}$ becomes more positive.

Hence, E_{cell} will be more positive.

- (c) Compound **A**, $\text{C}_{11}\text{H}_{15}\text{NO}_2$, exhibits geometric isomerism. It is insoluble in water but dissolves in hydrochloric acid to form a colourless solution. 1 mole of **A** reacts with Na to form 1 mole of H_2 gas. **A** produces a yellow precipitate when warmed with alkaline aqueous I_2 . **A** decolourises aqueous Br_2 , forming a white precipitate with the molecular formula of $\text{C}_{11}\text{H}_{15}\text{Br}_2\text{NO}_3$.

When **A** is treated with $\text{K}_2\text{Cr}_2\text{O}_7$ under suitable conditions, compound **B**, $\text{C}_{11}\text{H}_{11}\text{NO}_2$ is formed. Compound **B** forms a brick-red precipitate with Fehling's solution.

A reacts with hot aqueous KMnO_4 under suitable conditions to give **C**, $\text{C}_8\text{H}_7\text{NO}_4$, liberating a gas which gives a white precipitate with limewater.

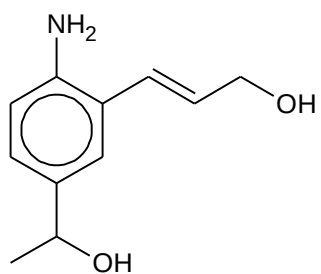
Deduce the structures of compounds **A**, **B** and **C**, giving reasons for your answer.

[10]

Information	Deductions
A has a high C:H ratio $\approx 1:1$, number of C > 6.	A contains a benzene ring
A exhibits geometric isomerism	A contains C=C bond/alkene and each C atom of the C=C double bond is bonded to 2 different groups of atoms.
A undergoes <u>acid-base</u> reaction with hydrochloric acid.	A contains an (basic) amine group
1 mol of A undergoes <u>redox</u> with Na to form 1 mol of H ₂ gas.	A contains <u>2</u> alcohol groups / phenol
A undergoes <u>oxidation</u> with I ₂ .	A contains CH ₃ CO or CH ₃ CH(OH)-group.
A undergoes electrophilic substitution and electrophilic addition with aqueous Br ₂ .	A contain a phenol / phenylamine and a C=C double bond/alkene. , There are 2 substituents on 2-,4- or 6- positions of the phenol group.
A undergoes <u>oxidation</u> with K ₂ Cr ₂ O ₇ with the loss of 4H atoms and no gain of O atom.	A contains 2 alcohol groups which undergo oxidation. B does not contain carboxylic acid.
B undergoes <u>oxidation</u> with Fehling's solution.	B contains an aliphatic aldehyde / not a benzaldehyde. A contains a 1° alcohol.
A undergoes <u>oxidation</u> with KMnO ₄ .	C contains 2 alkyl side chains which can undergo oxidation (since C has 8 C, it has 2 COOH substituents)

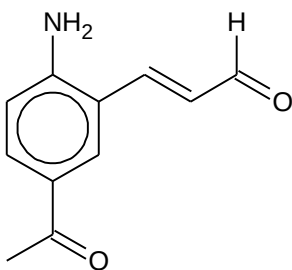
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A

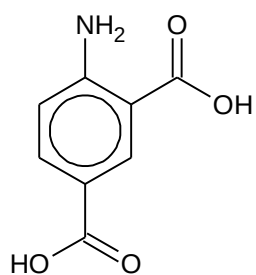


substituents can be in either
2,4 or 6 positions

B

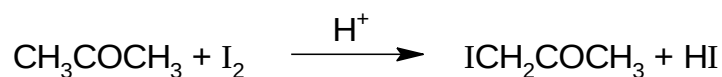


C



[Total: 20]

- 2 (a) Propanone reacts with iodine in the presence of an acid. The kinetics of the reaction



can be investigated experimentally by varying the concentration of the three substances involved and determining the time for the colour of the iodine to disappear. In this method, the rate at which the iodine concentration changes,

$$\text{rate of reaction} \propto \frac{\text{volume of aqueous iodine used}}{\text{time for colour of iodine to disappear}}$$

The following results were obtained in such an experiment.

experiment number	volume of 1 mol dm ⁻³ propanone / cm ³	volume of 0.01 mol dm ⁻³ iodine / cm ³	volume of 1 mol dm ⁻³ sulfuric acid / cm ³	volume of water / cm ³	relative time for iodine colour to disappear
1	8	4	8	0	1
2	8	4	4	4	2
3	4	4	8	4	2
4	8	2	8	2	0.5

- (i) Deduce the order of reaction with respect to propanone, iodine and hydrogen ions. [3]

$$\text{Let rate} = k[\text{propanone}]^x[\text{I}_2]^y[\text{H}^+]^z$$

Comparing expt 1 and 2,

$$\frac{4/1}{4/2} = \frac{k(8)^x(4)^y(8)^z}{k(8)^x(4)^y(4)^z}$$

$$z = 1$$

Comparing expt 1 and 3,

$$\frac{4/1}{4/2} = \frac{k(8)^x(4)^y(8)}{k(4)^x(4)^y(8)}$$

$$x = 1$$

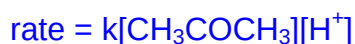
Comparing expt 1 and 4,

$$\frac{4/1}{4/0.5} = \frac{k(8)(4)^y(8)}{k(8)(2)^y(8)}$$

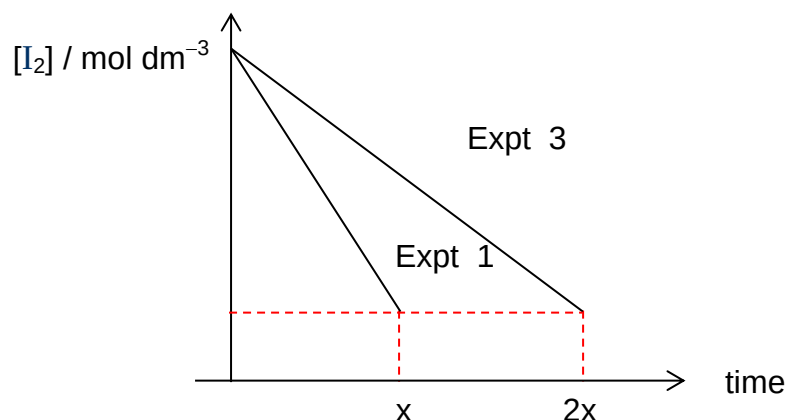
$$y = 0$$

The order of reaction w.r.t. to propanone, iodine and hydrogen ions are 1, 0 and 1 respectively.

- (ii) Write the rate equation for the reaction. [1]



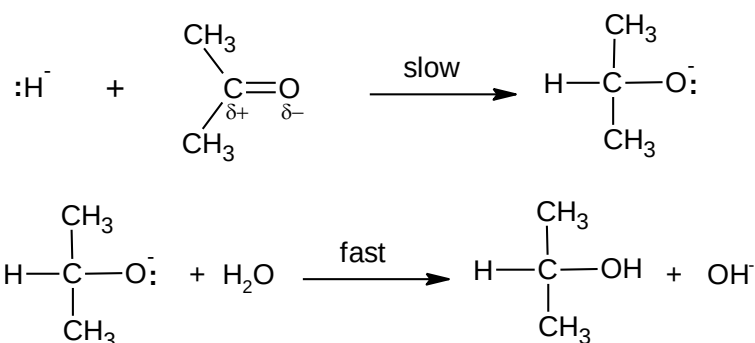
- (iii) Sketch a graph to show the change in concentration of I_2 with time for experiments 1 and 3. Label each experiment clearly on your graph. [2]



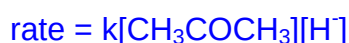
- (iv) The reaction between propanone and bromine proceeds by a similar mechanism. How would you expect the rate of this reaction to compare with that of the above reaction? Explain your answer. [1]

The rate remains the same because the halogen (bromine/ iodine) does not appear in the rate equation / the halogen is not involved in the rate determining step.

- (b) Kinetics studies suggests that the mechanism for the reduction of propanone by lithium aluminium hydride involves the following two steps:



- (i) Construct a rate equation for this reaction. [1]



- (ii) Suggest how the rate of the reaction of $\text{CH}_3\text{CH}_2\text{CHO}$ and LiAlH_4 compares with that of CH_3COCH_3 , giving your reasoning. [2]

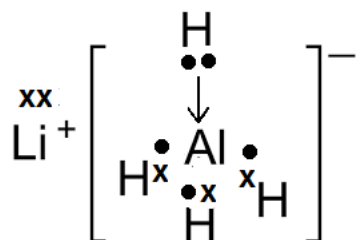
Rate of reaction of $\text{CH}_3\text{CH}_2\text{CHO}$ and LiAlH_4 would be faster than that of CH_3COCH_3 .

This is because the C of $\text{C}=\text{O}$ of aldehyde $\text{CH}_3\text{CH}_2\text{CHO}$ is more δ^+ as it has fewer electron donating alkyl groups bonded to it, making it more susceptible to nucleophilic attack.

Or

As $\text{CH}_3\text{CH}_2\text{CHO}$ has fewer alkyl groups attached to the $\text{C}=\text{O}$, there is less steric hindrance for the H^- to attack.

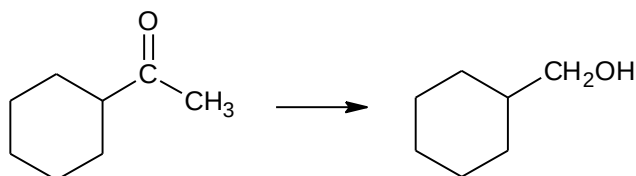
- (iii) Draw the 'dot and cross' diagram, showing the electrons (outer shells only) in lithium aluminium hydride, LiAlH_4 . In your diagram, use the symbol ' $\square$ ' for any additional electrons responsible for the overall negative charge of the anion. [2]

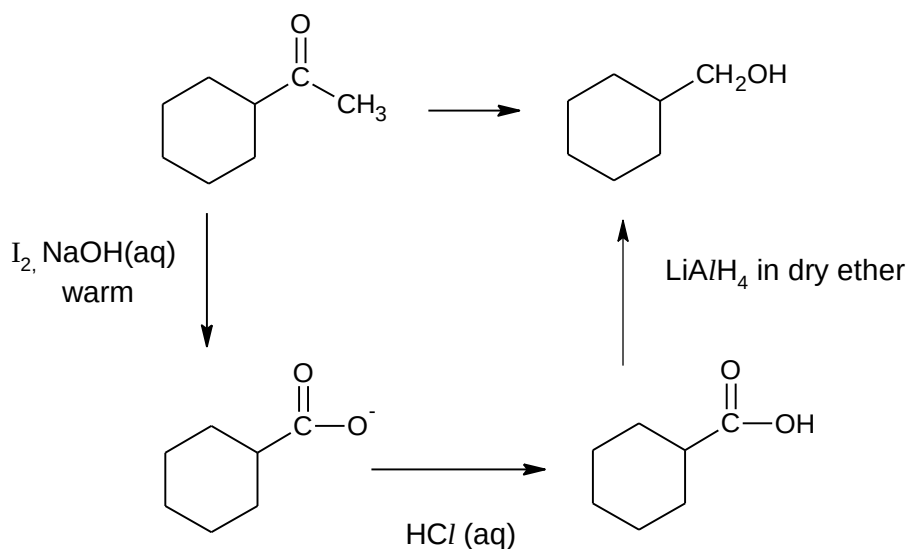


- (iv) Suggest the shape of the anion in LiAlH_4 . [1]

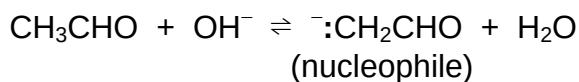
tetrahedral

- (c) In no more than 3 steps, outline how the following transformation can be achieved. State the reagents and conditions for each step, as well as the structures of any intermediates formed. [2]





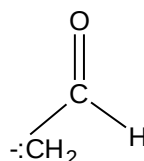
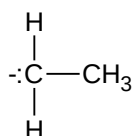
- (d) In the *aldol* reaction, a strong base (e.g. the OH^- ion) is used to generate a nucleophile from ethanal as shown in the following equation.



- (i) Explain why the nucleophile $\text{:CH}_2\text{CH}_3$ cannot be generated from the reaction of ethane with hydroxide ion. [2]

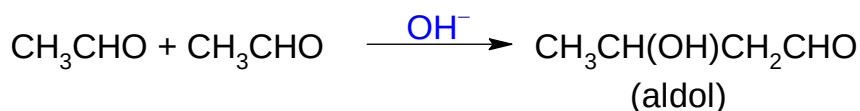
Anion from ethane

Anion from ethanal



Anion from aldehyde is more stable due to electron-withdrawing C=O group which helps to disperse the negative charge / the negative charge on O can be delocalised into the C=O group. The anion from ethane, however, is not stable as the negative charge is localised on the C atom. The presence of the electron donating CH_3 group intensifies the negative charge, destabilising the anion further.

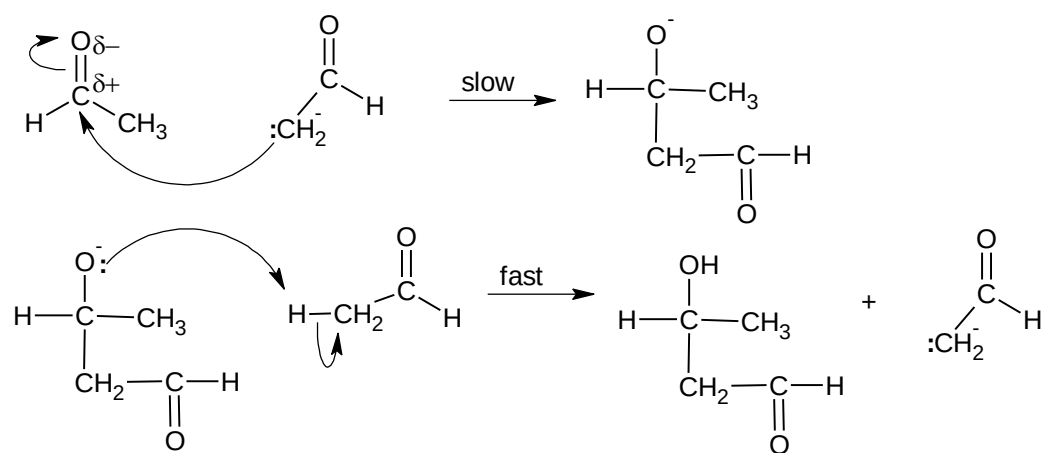
- (ii) By considering the nucleophile generated from ethanal and the equation for the *aldol* reaction,



name and describe the mechanism for the reaction.

[3]

Nucleophilic addition



[Total:20]

- 3 (a)** The water molecule can react in various ways: as an acid, as a base, as a nucleophile, as an oxidising reagent and as a reducing agent.

Study the following reactions and decide in which way water is reacting in each case. Explain your answers fully.



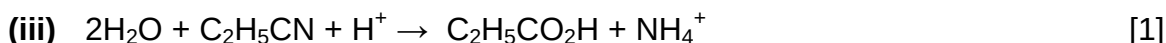
As an acid, H_2O donates a proton, H^+ , to H^- from NaH to form hydrogen and forms conjugate base, OH^- .



H_2O acts as a reducing agent, the oxidation number of F_2 decreased from 0 in F_2 to -1 in HF .

Or

H_2O acts as a reducing agent, as it is oxidised to O_2 with the oxidation number of O increased from -2 in H_2O to 0 in O_2 .



H_2O acts as a nucleophile attacking the δ^+ carbon to hydrolyse propanenitrile



H_2O acts as an oxidising agent, the oxidation number of Na increased from 0 in Na to +1 in NaOH .

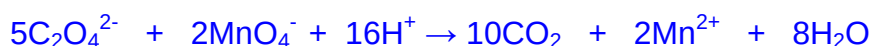
Or

As an oxidising agent, as H_2O is reduced to H_2 with the oxidation number of H decreased from +1 in H_2O to 0 in H_2 .

- (b)** Acidified potassium manganate, KMnO_4 , is a strong oxidising agent which can oxidise both X^{2+} and $\text{C}_2\text{O}_4^{2-}$ in XC_2O_4 . One of the products of this reaction is CO_2 .

37.50 cm^3 of $0.0200 \text{ mol dm}^{-3}$ acidified KMnO_4 was found to completely oxidise 25.0 cm^3 of $0.050 \text{ mol dm}^{-3}$ XC_2O_4 solution.

- (i)** Write a balanced equation for the reaction between MnO_4^- and $\text{C}_2\text{O}_4^{2-}$. Hence, determine the volume of KMnO_4 needed to oxidise only $\text{C}_2\text{O}_4^{2-}$. [2]



No. of moles of $\text{C}_2\text{O}_4^{2-}$ ions = $0.025 \times 0.1 = 0.00125 \text{ mol}$

No. of moles of MnO_4^- needed to oxidise $\text{C}_2\text{O}_4^{2-} = 0.00125 \times \frac{2}{5}$
 $= 5.00 \times 10^{-4} \text{ mol}$

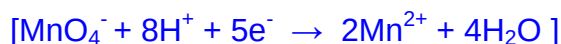
$$\text{Thus, volume of MnO}_4^- \text{ needed} = (5.00 \times 10^{-4} / 0.0200) \times 1000 \\ = 25 \text{ cm}^3$$

- (ii) Using your answer in (i), find the oxidation state of **X** in the product. [2]

$$\text{Volume of MnO}_4^- \text{ ions needed to oxidise X}^{2+} \text{ ions} = 37.50 - 25.00 \\ = 12.50 \text{ cm}^3$$

$$\text{No. of moles of MnO}_4^- \text{ ions} = 0.0125 \times 0.0200 \\ = 2.50 \times 10^{-4}$$

$$\text{No of moles of electrons accepted by MnO}_4^- = 2.50 \times 10^{-4} \times 5 \\ = 0.00125 \text{ mol}$$



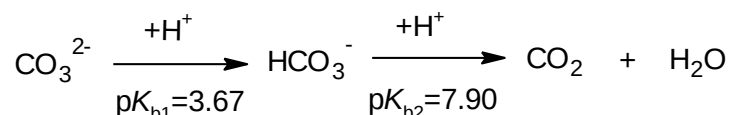
$$\text{No. of moles of X}^{2+} \text{ ions} = 0.00125 \text{ mol,}$$

$$\text{Mole ratio of X}^{2+} : \text{e} = 1 : 1$$

Since 1 mole of X^{2+} loses 1 mole of electrons,

Oxidation state of **X** in the product = +3

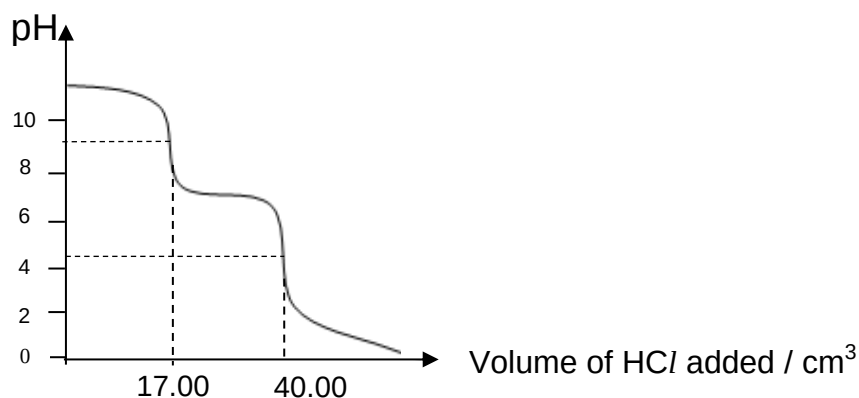
- (c) The CO_3^{2-} ion reacts with H^+ in two stages.



A solution contains a mixture of aqueous sodium carbonate, Na_2CO_3 and sodium hydrogencarbonate, NaHCO_3 .

25.0 cm^3 of the solution was titrated against 0.300 mol dm^{-3} hydrochloric acid, HCl using phenolphthalein as an indicator until a colour change was observed.

The titration was then continued by using a second indicator, methyl orange. The titration curve for the reaction is given below.



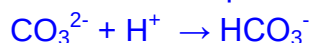
- (i) State the colour changes observed at the first and second end-points. [1]

1st end point: pink/ red to colourless

2nd end point: yellow to orange

- (ii) Calculate the concentration, in mol dm⁻³, of Na₂CO₃ in the 25.0 cm³ solution. [1]

For the 1st end point,



Amount of CO₃²⁻ in 25.0 cm³ of solution

$$= \frac{17.00}{1000} \times 0.300 = 5.1 \times 10^{-3} \text{ mol}$$

$$\text{Concentration of Na}_2\text{CO}_3 = \left(\frac{5.10 \times 10^{-3}}{\frac{25.0}{1000}} \right) = 0.204 \text{ mol dm}^{-3}$$

- (iii) From the graph, determine the volume of acid used to react with HCO₃⁻ originally present in the 25.0 cm³ solution.

Hence, determine the concentration of the HCO₃⁻ in the 25.0 cm³ solution. [2]

volume of acid used to react with HCO₃⁻ in the original solution
= 40 – 17 – 17 = 6 cm³

$$\begin{aligned} \text{Concentration of NaHCO}_3 &= \left(\frac{6.00}{1000} \times 0.300 \right) \times \frac{1000}{25.0} \\ &= 0.072 \text{ mol dm}^{-3} \end{aligned}$$

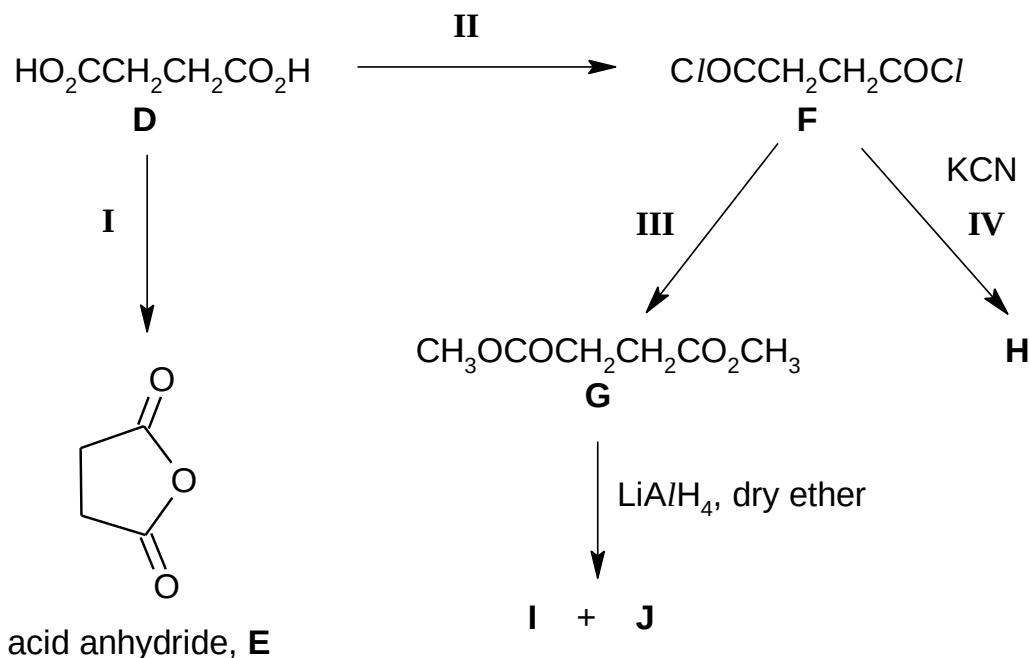
- (iv) Using your answer in (ii) and (iii) and any other information provided, calculate the pH of the original solution. [2]

$$\text{pOH} = \text{pK}_b + \lg \frac{[\text{salt}]}{[\text{base}]}$$

$$\text{pOH} = 3.67 + \lg \frac{(0.072)}{0.204} = 3.22$$

$$\text{pH} = 14 - 3.22 = 10.8 \text{ (3 s.f.)}$$

- (d) Consider the following reaction scheme involving butane-1,4-dioic acid.

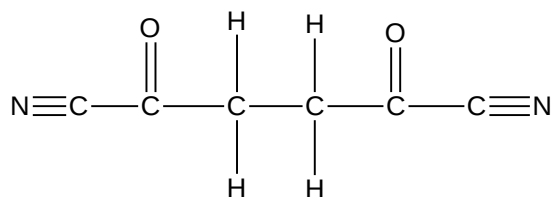


- (i) Suggest the reagent and condition for step II. [1]

step II: $\text{PCl}_5(\text{s})$, room temperature

- (ii) Draw the structures of the compounds, **H**, **I** and **J**. [3]

H:

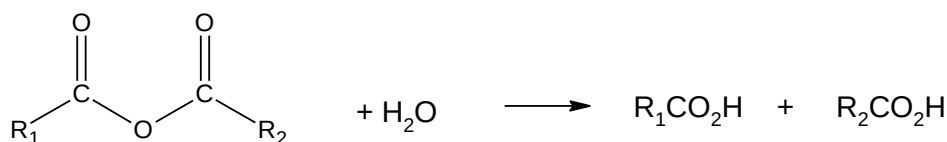


I and J: $\text{CH}_3\text{OH} + \text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

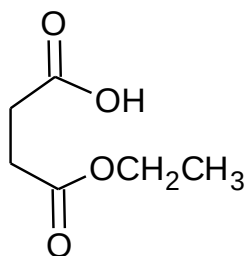
- (iii) What type of reaction has occurred in the formation of the acid anhydride **E** in step I? [1]

condensation / nucleophilic addition- elimination

- (iv) Acid anhydrides undergo the same reactions as acyl chlorides, but at a slower rate. Whilst acyl chlorides react with water to produce a carboxylic acid and HCl , acid anhydrides react with water to produce two carboxylic acids as shown below.



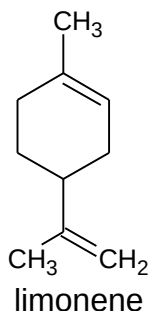
Suggest the structural formula of the product obtained when acid anhydride **E** reacts with ethanol instead of water. [1]



[Total: 20]

- 4 (a) Concentrated sulfuric acid reacts with solid potassium bromide to form two different bromine-containing products **K** and **L**.

Treating limonene, $C_{10}H_{16}$, separately with **K** and **L** produces compounds **M**, $C_{10}H_{18}Br_2$ and **N**, $C_{10}H_{16}Br_4$ respectively.

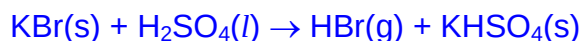


Separate reactions of compounds **M** and **N** with hot aqueous NaOH give **P**, $C_{10}H_{20}O_2$ and **Q**, $C_{10}H_{20}O_4$ respectively.

Q decolourises hot acidified purple $KMnO_4$, to yield **R**, $C_{10}H_{16}O_5$, but **P** does not.

- (i) By identifying the bromine-containing products **K** and **L**, write equations for their formation. [2]

substance **K**

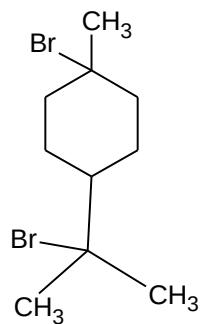


substance **L**

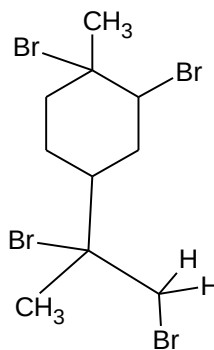


- (ii) Write an equation for the reaction between **L** and hot aqueous NaOH. [1]
 $3Br_2(g) + 6NaOH(aq) \rightarrow 5NaBr(aq) + NaBrO_3(aq) + 3H_2O(l)$

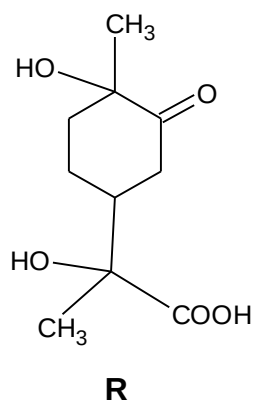
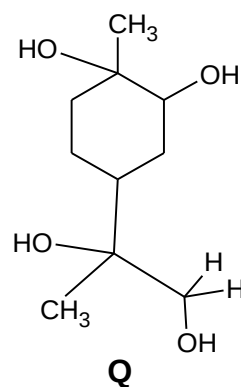
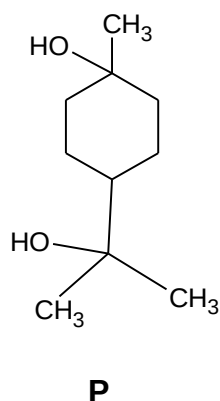
- (iii) Draw the structural formulae of compounds **M**, **N**, **P**, **Q** and **R**. [5]



M



N



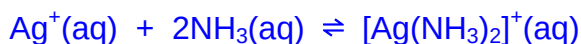
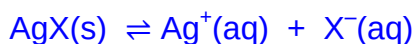
- (iv) By quoting appropriate data from the *Data Booklet*, explain how the rate of reaction with aqueous NaOH will change when the bromine atoms of compound **M** are replaced with iodine atoms? [2]

Bond energies: C – I = +240 kJ mol⁻¹; C – Br = +280 kJ mol⁻¹

As the rate determining step involves the breaking of the C-Hal bond, the C- I bond being longer and weaker is easier to break, hence the rate of reaction with the iodoalkane will be faster.

- (b) Describe what is observed when each of the halide ions, Cl⁻, Br⁻ and I⁻ reacts with aqueous silver nitrate, followed by aqueous ammonia (dilute and concentrated). [5]

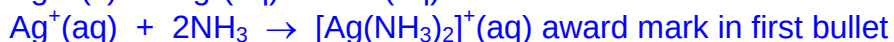
	Cl ⁻ (aq)	Br ⁻ (aq)	I ⁻ (aq)
AgNO ₃ (aq)	White ppt of AgCl	Cream ppt of AgBr	Yellow ppt of AgI
Solubility of AgX in			
(i) excess dilute aqueous NH ₃	soluble	Insoluble	Insoluble
(ii) excess concentrated aqueous NH ₃	soluble	Soluble	Insoluble



For AgCl:

The formation of diamminesilver(I) complex in dilute aqueous ammonia is sufficient to cause the ionic product of AgCl to fall below its K_{sp} value (as the K_{sp} value of AgCl is highest). Thus, AgCl(s) is soluble in both dilute and concentrated $\text{NH}_3(\text{aq})$.

For AgBr:



The formation of diamminesilver(I) complex in dilute aqueous ammonia is not sufficient to cause the ionic product of AgBr to fall below its K_{sp} value.

However, with concentrated aqueous ammonia, the formation of diamminesilver(I) complex is sufficient to decrease the $[\text{Ag}^+]$, thus causing the ionic product of AgBr to fall below its K_{sp} value thus ppt dissolves. This explains why AgBr(s) is soluble in concentrated $\text{NH}_3(\text{aq})$ but insoluble in dilute $\text{NH}_3(\text{aq})$.

For AgI:

The formation of diamminesilver(I) complex in both dilute and concentrated aqueous ammonia is not sufficient to cause the ionic product of AgI to fall below its K_{sp} value (as the K_{sp} value of AgI is extremely low). Therefore, AgI(s) is insoluble in both dilute and concentrated $\text{NH}_3(\text{aq})$.

- (c) Use the *Data Booklet* to describe the relative reactivity of the halogens as oxidising agents and relate this reactivity to the relevant $E^\ominus$ values. [2]



The oxidising power of halogen decreases down the group as shown by the less positive standard electrode potential.

- (d) The halogens form a number of interhalogen compounds, including the pale yellow liquid, BrF_3 . This compound is a powerful fluorinating agent, converting silicon dioxide into silicon tetrafluoride and two other elements as products.

- (i) Predict the shape of BrF_3 . [1]

T- shaped

- (ii) Suggest a balanced equation for the reaction. [1]



- (iii) Use your equation to calculate the mass of BrF_3 needed to react with 10.0 g of silicon dioxide. [1]

$$\begin{aligned}\text{Amount of SiO}_2 \text{ used} &= 10.0 / [(28.1 + 2(16.0))] \\ &= 0.1663 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Mass of BrF}_3 &= (4/3)(0.1663)[79.9 + 3(19.0)] \\ &= 30.4 \text{ g}\end{aligned}$$

[Total: 20]

- 5 (a) (i) The melting point of SO_2 is -72°C while that of Al_2O_3 is 2072°C . Account for the difference in their melting points, in terms of their structure and bonding. [3]

Al_2O_3 has giant ionic lattice structure whereas SO_2 has simple molecular / simple covalent structure. Melting Al_2O_3 involves breaking strong ionic bonds between Al^{3+} and O^{2-} ions which require more energy, compared to overcoming weak intermolecular permanent dipole-dipole forces between SO_2 molecules. Hence Al_2O_3 has a much higher melting point.

- (ii) MgO , Al_2O_3 and SiO_2 exist as white solids. If a sample of one of the oxides was provided as a white powder, describe the reactions you could carry out on the powder to determine which of the three oxides it was. [3]

Divide the sample into two portions in separate test tubes.

To the first portion, add dilute hydrochloric acid.

To the second portion, add dilute sodium hydroxide.

Note the solubility of the sample in each case.

If the sample is soluble in dilute hydrochloric acid, but insoluble in dilute sodium hydroxide, then it is MgO .

If the sample is soluble in dilute hydrochloric acid and dilute sodium hydroxide, then it is Al_2O_3 .

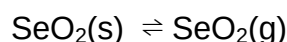
If the sample is insoluble in dilute hydrochloric acid and dilute sodium hydroxide, it is SiO_2 .

Or Table:

	MgO	Al_2O_3	SiO_2
Add dilute hydrochloric acid	Soluble	Soluble	Insoluble
Add dilute NaOH	Insoluble	Soluble	Insoluble

(Note: All three white solids are insoluble in water.)

- (b) (i) When solid selenium dioxide undergoes sublimation, an equilibrium exists between the two phases.



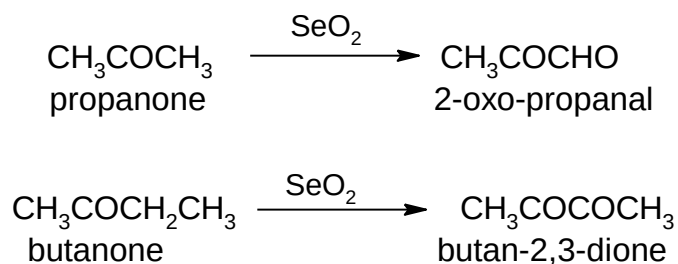
The standard enthalpy change, $\Delta H^\ominus$, and the standard entropy change, $\Delta S^\ominus$, of sublimation are $+111 \text{ kJ mol}^{-1}$ and $+183 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively. Predict whether the sublimation of solid selenium dioxide is spontaneous at room temperature. [2]

$$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$$

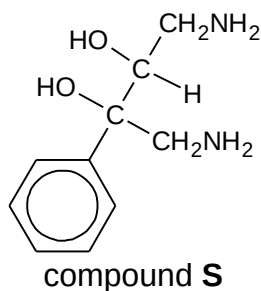
$$\Delta G^\ominus = (+111) - (298)(+183/1000) = +56.5 \text{ kJ mol}^{-1} > 0, \text{ hence it is not}$$

spontaneous.

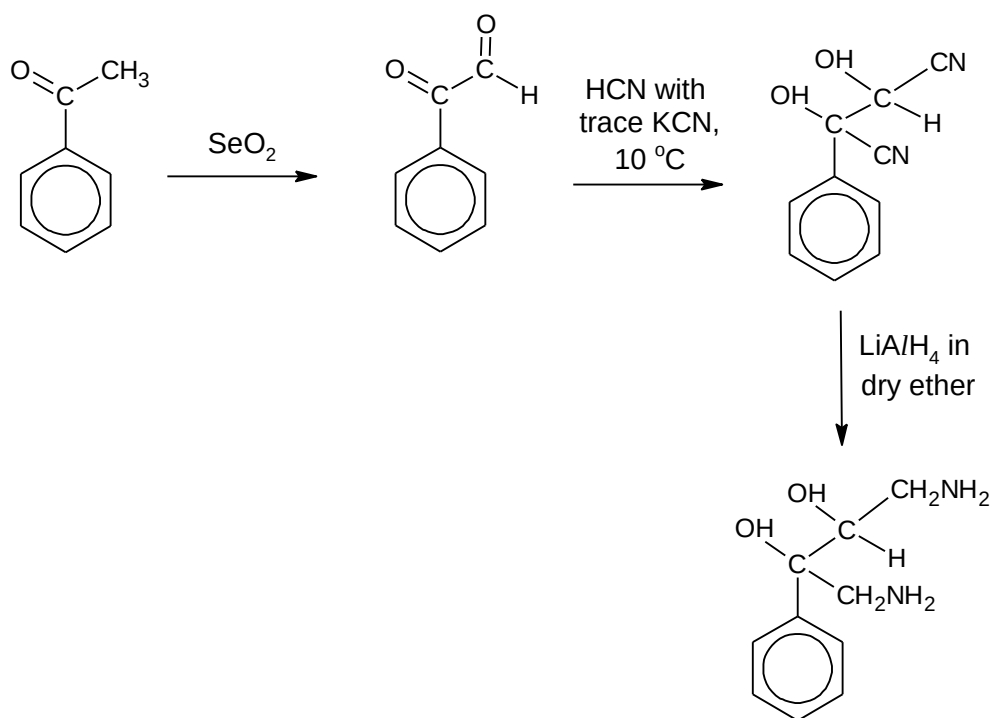
- (ii) In 1932, Riley and co-workers published their findings on the use of SeO_2 to introduce carbonyl functional group into ketones. The following are some examples of the Riley oxidation.



Starting from 1-phenylethanone, devise a three-step synthesis for compound **S**. You are to use Riley oxidation in one of your steps. For each step, give the reagents and conditions and give the structures of the intermediates formed.



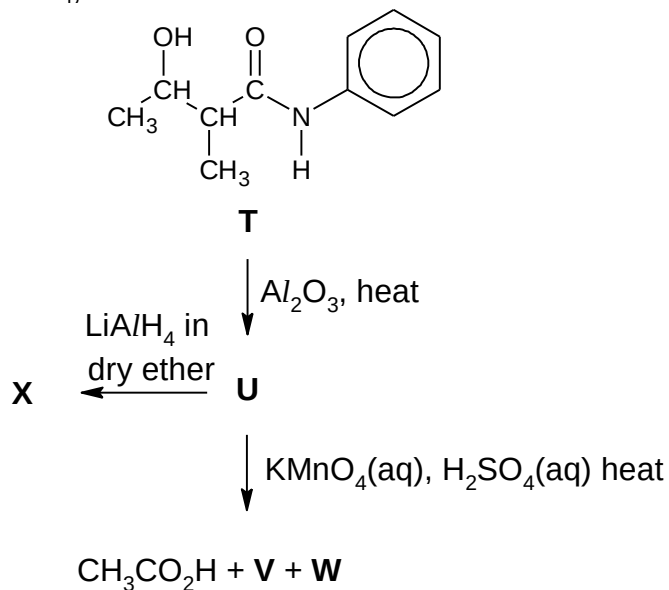
[3]



(c) Compound **T** undergoes dehydration on heating with Al_2O_3 to form compound **U**.

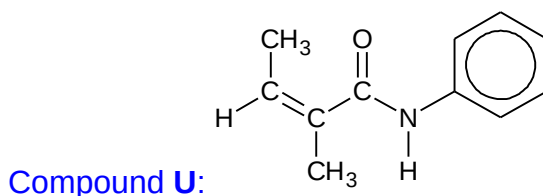
U undergoes reaction with hot acidified KMnO_4 , to form three products, **V**, **W** and ethanoic acid. **V** was found to give a yellow precipitate with alkaline aqueous iodine.

On treatment with LiAlH_4 , **U** forms **X**.

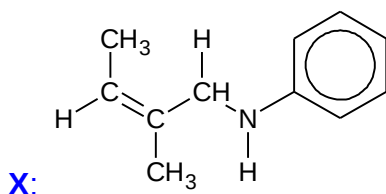
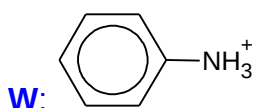


(i) Give the structures of **U**, **V**, **W** and **X**.

[4]



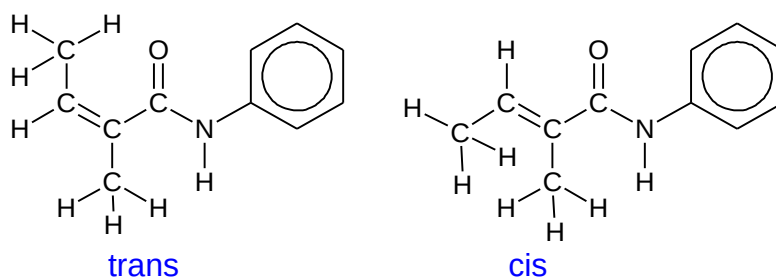
V: $\text{CH}_3\text{COCO}_2\text{H}$



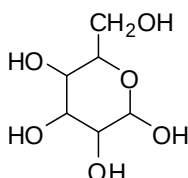
(ii) What type of stereoisomerism can **U** exhibit? Draw the displayed formulae of the stereoisomers of **U**.

[2]

Cis-trans / geometric isomerism

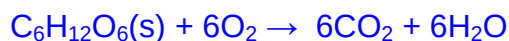


- (d) Glucose, an important simple sugar or monosaccharide, is required by the cells in our body as a source of energy and metabolic intermediate.



Glucose, $C_6H_{12}O_6$, undergoes complete combustion to form carbon dioxide and water.

- (i) Using the molecular formula of glucose, write a balanced equation for the complete combustion of glucose. [1]



- (ii) Use bond energy values from the *Data Booklet* to calculate the enthalpy change of combustion of glucose. [Rather than the $C=O$ value given in the *Data Booklet*, use a value of 805 kJ mol^{-1} for the bond energy of each $C=O$ bond in CO_2 .] [2]

Bond energies

Bonds broken	Bond energy / kJ mol^{-1}
C – C	5 x 350
C – O	7 x 360
O – H	5 x 460
C – H	7 x 410
O = O	6 x 496

Bonds formed	Bond energy / kJ mol^{-1}
C=O	12 x 805
O – H	12 x 460

$$\begin{aligned}
 \Delta H &= +\sum m\text{B.E.}(\text{reactants}) - \sum n\text{B.E.}(\text{products}) \\
 &= +[5\text{BE}(\text{C-C}) + 7\text{BE}(\text{C-O}) + 5\text{BE}(\text{O-H}) + 7\text{BE}(\text{C-H}) + 6 \text{BE}(\text{O=O})] - \\
 &\quad [12\text{BE}(\text{C=O}) + 12\text{BE}(\text{O-H})] \\
 &= -2764 \text{ kJ mol}^{-1} = -2760 \text{ kJ mol}^{-1}
 \end{aligned}$$

[Total: 20]

End of Paper